# Validation Noise-Bank Subtraction and an Event-Conditioned TF-DPCRN Ensemble for Noise Event Removal in Vaani Speech

Shah Ahmed Shakir Abu Asim Khan*, Sajid Javid†, Osama Zaheer‡, Ahraz Shamim§
*Aligarh Muslim University †IIIT Delhi ‡Binary Semantics §Vecmocon Technologies
Team CodeAMU, IndoML 2026 Datathon, Track 2 Contact: sajidj@iiitd.ac.in

***Abstract*—Track 2 of the IndoML 2026 Datathon asks for the removal of non-speech noise events from Vaani speech recordings, scored by SI-SDR on synthetic clips plus the reduction in word error rate of the mandated SraVaani-1.0 recogniser. Our system has two parts. (i) A three-member TF-DPCRN ensemble, a complex-mask network with dual-path recurrent and attention blocks conditioned on noise-event windows from our Track 1 detector, trained from scratch on Vaani speech, labelled noise events and the organisers' validation pairs, then fine-tuned on mixtures that follow the measured Phase 2 test statistics. (ii) Noise-bank subtraction: the synthetic test clips reuse noise recordings that the validation release exposes (mixture minus clean reference), so we locate them by cross-correlation, subtract them with a least-squares gain, and let the enhancer clean the residual left by the organisers' slight alteration of the copies, a procedure the organisers approved in writing. Team CodeAMU ranked 1st on the final (Phase 2) Track 2 leaderboard with Combined 26.4894 (SI-SDR 20.34 dB, $\Delta$WER 6.15 points), against 15.67 for the same enhancer without the bank.**

***Index Terms*—speech enhancement, noise event removal, DPCRN, complex masking, cross-correlation, Vaani**

## I. Introduction

For each noisy test clip (Phase 2: 3,178 clips, 10.9 h, 16 kHz) a system must return an enhanced waveform of the same length and a transcript of it produced by the mandated ASR model SraVaani-1.0 [1]. The score is Combined = SI-SDR [2] (dB, averaged over the organiser-synthesised clips) + $100 \cdot \Delta$WER, where $\Delta$WER is the WER reduction over all clips. The synthetic clips add noise events to clean speech; natural clips contain real noise and no reference.

*Final standing:* **1st place** on the Phase 2 final leaderboard with Combined **26.4894**: SI-SDR 20.34 dB, $\Delta$WER 6.15 points (18.09 on synthetic and 0.52 on natural clips), transcript audit passed. Training and inference code, configurations, the enhancer checkpoints and the noise bank: https://github.com/sajidjavid222/CodeAMU_Track2.

## II. Training Data and Preprocessing

### A. Vaani data

Speech beds come from the Vaani corpus [3], [4]: 14,562 utterances (21.0 h, 35 districts) selected by duration (2–16 s), estimated SNR, clipping and voice-activity ratio (WebRTC VAD [6] $\geq 0.35$); median estimated SNR 32 dB. From the gold tier of the Vaani Noise Event Dataset [5] we take 459 labelled noise events (0.12–8 s, at most 55% speech), 2,900 event-free clean segments (2.9 h) and 75 noise floors. Audio is 16 kHz mono, processed in 6 s (base) or 8 s (fine-tuning) crops.

### B. Official validation release

Each of the 695 synthetic validation clips ships its clean reference, so *noise* = *mixture* − *clean*; the mixing is additive and confined to the labelled spans. Under 30.3% of the labelled spans the clean reference is digital silence (pure noise, the ideal output is silence); under 56.4% speech is present at a median local SNR of $-7.3$ dB. We split the clips by file-name group into a training fold (557 pairs) and a held-out fold (138 pairs whose clean speech is never used for training; their noise recordings, like every pool recording, reach the fine-tuning mixer through the noise bank). The training fold gives the organiser pairs themselves (0.79 h), 2,164 noise bursts and 530 noise floors; 374 further floors are harvested from at most 400 natural validation clips (noisy minus enhanced, outside the noise windows). The full release also provides the noise bank (Sec. III-D). No other data is used.

### C. Mixture generation and augmentation

*Base mixer.* With probability 0.65 a batch item is a randomly cropped organiser training pair (gain $\pm 6$ dB, noise re-mixed within the fold); otherwise a bed (Vaani, gold speech or gold clean) receives a noise floor (SNR drawn from the distribution measured on the organiser pairs, slow amplitude modulation with probability 0.5) and 0–4 noise events (count, duration and local SNR drawn from the measured distributions widened by 35%; 12% of clips without events; overlaps allowed; 1–80 ms fades), with random exponential-decay reverberation (RT 0.15–0.7 s, probability 0.35) and channel EQ (probability 0.5). A polish stage lowers the organiser share to 0.6. *Phase 2 mixer.* Measured with the noise bank on 100 confidently synthetic Phase 2 clips, the Phase 2 generator differs from Phase 1: 10.1% instead of 25.5% of events over digital silence, louder noise under speech (local SNR median $-9.7$ instead of $-7.6$ dB), 0.47 events/s and slightly altered noise copies (Table II). The fine-tuning mixer reproduces this: beds from organiser clean references (50%), Vaani speech (25%) and gold clean speech (25%); noise from the bank recordings (70%) and the event pool (30%), each altered by a random 3-tap FIR tilt, a $\pm 0.4\%$ resampling (probability 0.5) and

$-35$ dB hiss; events at 0.47/s, local SNR $\mathcal{N}(-9.7, 4)$ dB, 10% over silence, mixtures at $-22.8$ dBFS. In the last two stages organiser training pairs make up 30–50% (p2mix) and 40% (asr) of each batch. *Window augmentation.* The event-indicator input is perturbed during fine-tuning: all windows dropped (10%), single windows dropped (10%), edges jittered by up to 80 ms, false windows added (20%), 0.1 s padding (base stage: 35%, 25%, 150 ms, 20%).

## III. Proposed Method

### A. TF-DPCRN enhancer

Each member (3.9 M parameters, trained from scratch, no pretrained weights) takes the STFT $X$ (512-point Hann window, hop 128) as four channels: the real and imaginary parts of the power-compressed spectrum $X|X|^{-0.7}$ [7], the compressed magnitude $|X|^{0.3}$, and a per-frame noise-event indicator from the Track 1 event windows. An encoder of four $3\times3$ convolutions (stride 2 along frequency; 48/64/96/128 channels; batch norm, PReLU) feeds five dual-path blocks [8], [9]: a frequency BiLSTM and a time BiLSTM (128 units, each with a linear projection, layer norm and residual), followed by 4-head self-attention over time on the frequency-pooled features [10]. A transposed-convolution decoder with skip connections outputs a complex ratio mask [11], [12] with $\tanh$-bounded magnitude, $\hat{S} = \tanh(|M|)\, e^{j\angle M} X$. Complex masking matters here: on the synthetic validation clips an oracle magnitude mask (ideal ratio mask) reaches only 17.7 dB SI-SDR, because it keeps the mixture phase, which the noise dominates where speech lies at $-7$ dB local SNR.

### B. Loss and training strategy

$$\mathcal{L} = -\text{SI-SDR}(\hat{s}, s) + 0.1\,\mathcal{L}_{\text{MR}} + 0.3\,\mathcal{L}_{\text{mel}}, \qquad (1)$$

where $\mathcal{L}_{\text{MR}}$ averages spectral-convergence and log-magnitude losses [13] over STFT resolutions (512, 128), (1024, 256) and (256, 64), and $\mathcal{L}_{\text{mel}}$ is the L1 distance between 128-band log-mel spectra (25 ms window, 10 ms hop), a term aimed at the recogniser (weight 1.2 in the last stage, asr). The SI-SDR term skips items whose clean reference is silent. Optimisation: AdamW [14] (weight decay $10^{-4}$, betas 0.9/0.98), 1,000 (base) or 300 (fine-tuning) warm-up steps, cosine decay to 3% [15] over the stage's time budget, gradient clipping at 5, bf16, exponential moving average of the weights; checkpoints are evaluated every 10–15 minutes. Table I lists the stages; each member was trained within 6-hour jobs on one 71 GB MIG slice of an NVIDIA H200.

### C. Inference

The event windows are the Track 1 predictions of our detector ensemble [17] (24 detectors, combined audio and detector-output routing), padded by 0.1 s. The three members' waveforms are averaged after aligning their polarity with the mixture (SI-SDR is sign-blind, and unaligned averaging of two good members once gave 4.1 dB). The output is $y = \alpha\hat{s} + (1-\alpha)x$ with $\alpha = 1$ inside the windows and 0.9 outside, rescaled to the input RMS (which matters for the recogniser, not for SI-SDR) and peak-limited to 0.99. SraVaani-1.0, unchanged, transcribes the written 16-bit WAV files.

TABLE I

Training stages of each member and held-out SI-SDR (dB) of the 3-member ensemble after each stage. off0: 138 held-out organiser pairs (input 0.47 dB); p2h: 120 Phase 2-like mixtures from held-out speech (input $-5.2$ dB). Checkpoint selection (Sel.): off0, p2h, or fit = 1.53 off0 + 0.27 p2h, a fit of our leaderboard history.

| Stage | Data | min | lr | Sel. | off0 | p2h |
|---|---|---|---|---|---|---|
| base | base mixer + polish | 185+25 | $10^{-3}$ | off0 | 23.3 | 13.3 |
| p2ft | Phase 2 mixer | 40 | $2 \cdot 10^{-4}$ | p2h | 23.0 | 13.9 |
| cont | Phase 2 mixer | 45 | $10^{-4}$ | p2h | 23.0 | 14.0 |
| long | Phase 2 mixer | 300 | $10^{-4}$ | p2h | 23.5 | 14.7 |
| p2mix | + organiser pairs | 240 | $10^{-4}$ | fit | 24.1 | 14.9 |
| asr | + log-mel weight 1.2 | 40 | $10^{-4}$ | fit | 24.2 | 14.9 |

### D. Noise-bank subtraction (disclosure)

The organisers' synthetic clips paste whole noise recordings from a finite pool. In the validation release, *mixture* $-$ *clean* at the labelled timestamps gives 2,213 events, i.e. 1,181 distinct recordings after de-duplication. On clips that the Track 1 routers call synthetic (1,635 of 3,178), we pre-emphasise (0.97), resample to 8 kHz and cross-correlate the clip with every unit-norm template by FFT, normalised by the local signal energy [16]; up to three peaks per template with correlation $\geq 0.20$ are kept and matches are picked greedily by explained energy (overlap $\leq 30\%$). Each match with correlation $\geq 0.45$, strongest first, is refined by $\pm32$ samples at 16 kHz and subtracted as $g\,t$, where $t$ is the raw 16 kHz recording and $g \geq 0$ the least-squares gain on twice pre-emphasised signals: 9,065 copies in 1,635 clips. Phase 2 copies are altered by about $-26$ dB relative to the validation originals (the residual over silence rises from $-58$ to $-26$ dB; a per-copy FIR fit removes nothing, so the change is additive), so subtraction leaves a residual. The enhancer therefore runs again on the subtracted clip over the subtracted spans $\pm0.1$ s and the Track 1 windows, with $\alpha = 0$ outside them. The organisers confirmed in writing that noise components located in test audio by cross-correlation with validation-derived noise waveforms may be subtracted from the mixture in Track 2, provided the method is disclosed, as done here.

### E. Pretrained models and implementation

The enhancer uses no pretrained weights. External models: SraVaani-1.0 [1] (ARTPARK-IISc/SraVaani-1.0, revision f5dd535), the mandated recogniser, used unchanged for the transcripts and for local $\Delta$WER checks; and our Track 1 detectors and routers [17] for the event windows and the routing (their Whisper, WavLM and BEATs encoders are listed in that report). PyTorch 2.11 [18], CUDA 12.8, NVIDIA H200 (71 GB MIG slices). Phase 2 inference takes about 40 min on one slice: bank matching 18 min, subtraction 1 min, enhancement and ASR 19 min.

TABLE II
SYNTHETIC TEST CLIPS OF THE TWO PHASES, MEASURED WITH THE NOISE BANK ON 100 CONFIDENTLY SYNTHETIC CLIPS PER PHASE.

| | Phase 1 | Phase 2 |
|---|---|---|
| Median clip length (s) | 6.1 | 12.7 |
| Noise events per clip | 3.7 | 6.0 |
| Events over digital silence | 25.5% | 10.1% |
| Local SNR of events over speech, median (dB) | −7.6 | −9.7 |
| Residual after subtracting the matched copy (dB) | −58 | −26 |
| Best match correlation | 1.000 | 0.998 |

TABLE III
PHASE 2 LEADERBOARD PROGRESSION (3,178 TEST CLIPS). SI-SDR IN DB ON SYNTHETIC CLIPS; ΔWER IN WER POINTS OVER ALL CLIPS; "–": COMPONENT NOT RECORDED.

| Submission | SI-SDR | ΔWER | Comb. |
|---|---|---|---|
| Base ensemble (Phase 1 recipe) | 12.13 | 1.43 | 13.56 |
| + Phase 2 fine-tuning (p2ft) | 12.46 | 1.68 | 14.14 |
| + continued (cont) | 12.56 | 1.74 | 14.30 |
| + long fine-tuning (long) | 12.92 | 2.28 | 15.20 |
| + organiser pairs (p2mix) | – | – | 15.64 |
| + log-mel weight 1.2 (asr) | 13.38 | 2.29 | 15.67 |
| Bank subtraction only (p2mix on other clips) | – | – | 19.53 |
| Subtraction + enhancer pass: ASR-loss members | 20.10 | 6.16 | 26.27 |
| Subtraction + enhancer pass: p2mix | – | – | 26.44 |
| Subtraction + enhancer pass: p2mix + asr (5 models) | – | – | 26.46 |
| **Subtraction + enhancer pass: asr (final, 1st)** | 20.34 | 6.15 | **26.49** |

## IV. RESULTS AND CONCLUSION

Table III traces the Phase 2 leaderboard. The base ensemble, trained on Phase 1-style mixtures, fell from SI-SDR 17.46 dB on the Phase 1 test (Combined 18.83) to 12.13 dB on Phase 2. Fine-tuning on mixtures with the measured Phase 2 statistics recovered +1.64 Combined in four stages (13.56 to 15.20), and mixing real organiser pairs into the batches added +0.44 (15.64); 40 more minutes with the log-mel weight raised to 1.2 added +0.03 (15.67). On the 138 held-out organiser pairs the final ensemble reaches 24.2 dB, against 23.3 dB for the base ensemble (Table I). Local scores overshoot the leaderboard, and models trained from scratch on our own mixer overfitted it: an ensemble that gained on our Phase 2-like set but lost on the organiser pairs dropped 0.58 on the leaderboard, so we kept only fine-tunes that improved on the organiser pairs.

In Phase 1 the copies were exact and subtraction alone scored 53.24 (SI-SDR 49.97 dB). The Phase 2 alteration cut subtraction alone to 19.53. Running the enhancer over the subtracted spans restores most of the gap: in simulation on the held-out pairs with −26 dB alterations, SI-SDR rises from 26.5 dB (subtraction) to 31.4 dB (subtraction + enhancer), and on the leaderboard Combined rises from 19.53 to 26.44 with the p2mix ensemble. Members trained with an additional loss on SraVaani's encoder features raised ΔWER on model-only submissions (16.18 vs 15.64) but lost SI-SDR on the nearly clean subtracted audio (26.27). The milder log-mel continuation (asr) did best, 26.49 (SI-SDR 20.34 dB, ΔWER 6.15), ahead of the five-model union of p2mix and asr (26.46), and is our final system.

In conclusion, a compact event-conditioned complex-mask ensemble adapted to the measured test distribution, combined with exact subtraction of validation-derived noise copies and an enhancer pass over the residual, gave the winning Track 2 system (1st place, Combined 26.4894). The model-only path (15.67) remains the fallback for noise outside the validation pool.

*Reproducibility.* Run end to end from the released repository on the Phase 2 audio, the pipeline regenerates each bank-based submission exactly (asr, p2mix and the five-model union): all 3,178 WAV files are bit-identical to the submitted ones and all 3,178 transcripts are identical. The Track 1 repository regenerates the two inputs, event windows and routing, byte for byte from the cached detector outputs.

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